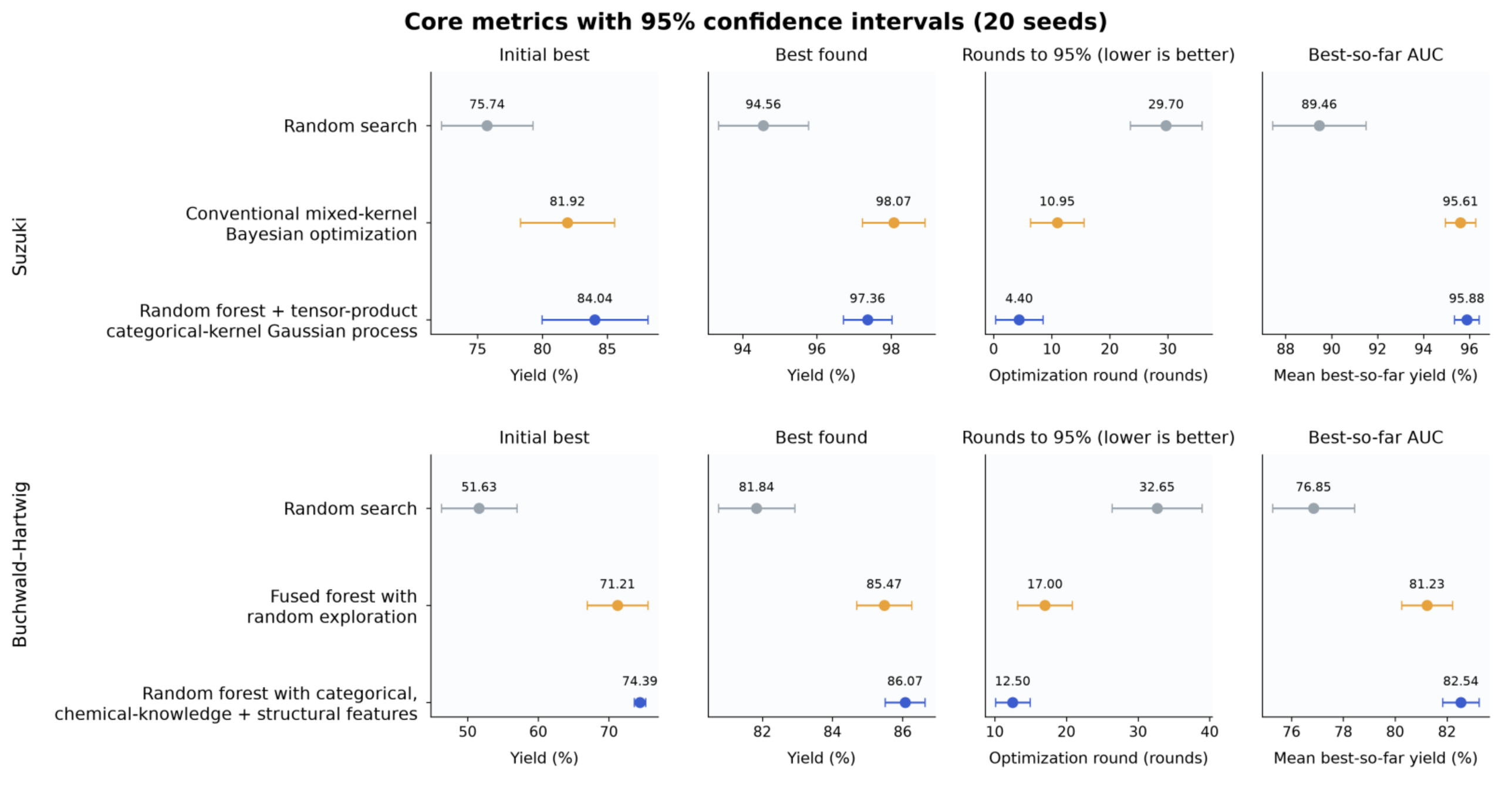

Core metrics with 95% confidence intervals (20 seeds)
Initial best
Best found
Rounds to 95% (lower is better)
Best-so-far AUC
Suzuki
Random search
Conventional mixed-kernel Bayesian optimization
Random forest + tensor-product categorical-kernel Gaussian process
75.74
81.92
84.04
94.56
98.07
97.36
29.70
10.95
4.40
89.46
95.61
95.88
75
80
85
94
96
98
0
10
20
30
88
90
92
94
96
Yield (%)
Yield (%)
Optimization round (rounds)
Mean best-so-far yield (%)
Buchwald–Hartwig
Random search
Fused forest with random exploration
Random forest with categorical, chemical-knowledge + structural features
51.63
71.21
74.39
81.84
85.47
86.07
32.65
17.00
12.50
76.85
81.23
82.54
50
60
70
82
84
86
10
20
30
40
76
78
80
82
Yield (%)
Yield (%)
Optimization round (rounds)
Mean best-so-far yield (%)